

TMS320C6748 LCDK

Kit Contents:

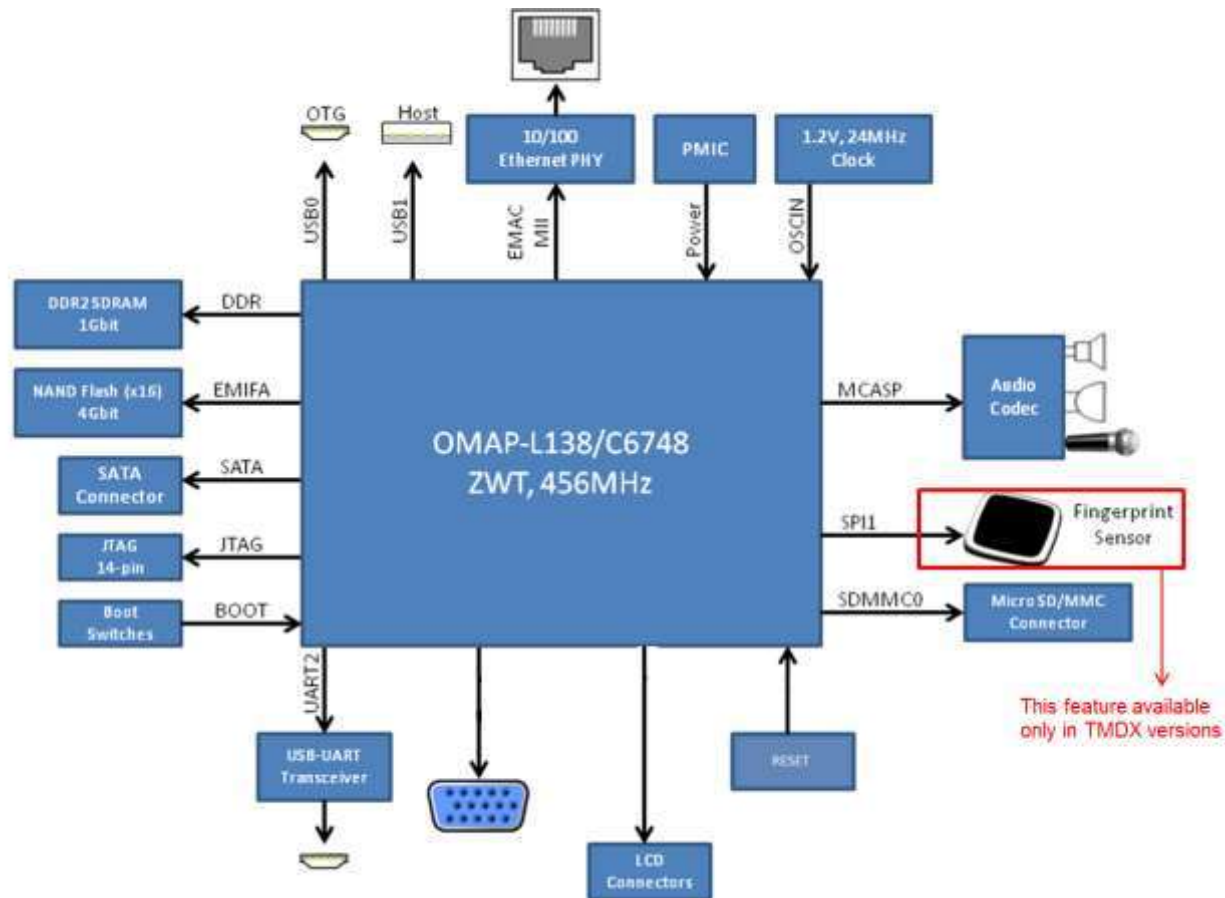
- TMS320C6748 Hardware Board
- Power Card(5VDC)
- XDS 100v2/3JTAG Emulator
- USB Cable
- User Manual & Example Programs(Soft Copy)



Description:

The TMS320C6748 DSP Starter kit (LCDK) is a low cost development tool development for practically learning Theoretical Concepts of Digital Signal Processing. DSP Algorithms used in Applications such as Speech Processing, Audio Processing, Image Processing and Digital Communication can be explored practically. The low-cost DSP Kit will also speed and ease development of real-time DSP programs with the help of Code Composer Studio Software A complete Integrated Development Environment (IDE), an efficient optimizing C/C++ compiler assembler, linker, debugger, and an advanced editor with Code Maestro™ technology for faster code creation, data visualization, a profiler and a flexible project manager

The TMS320C6748 DSP Starter Kit does not have an onboard emulator. An external emulator from TI (such as the XDS100, XDS200, XDS510, XDS560) or a third-party will be required to start development.



The C674x DSP core of C6000 processor family is a low-power applications processor and it consumes less power than other members of the TMS320C6000™ platform of DSPs

The DSP core uses a 2-level cache-based architecture. The level 1 program cache (L1P) is a 32- KB direct mapped cache, and the level 1 data cache (L1D) is a 32-KB 2-way, set-associative cache. The level 2 program cache (L2P) consists of a 256-KB memory space that is shared between program and data space. L2 memory can be configured as mapped memory, cache, or combinations of the two. Although the DSP L2 is accessible by the ARM9 and other hosts in the system, an additional 128KB of RAM shared memory is available for use by other hosts without affecting DSP performance.

Processor

- TI TMS320C6748 DSP Application Processor
- 456-MHz C674x Fixed/Floating Point DSP
- On-Chip RTC

Memory

- 128 MByte DDR2 SDRAM
- 128 MByte 16-bit wide NAND FLASH
- 1 Micro SD/MMC Slot

Interfaces

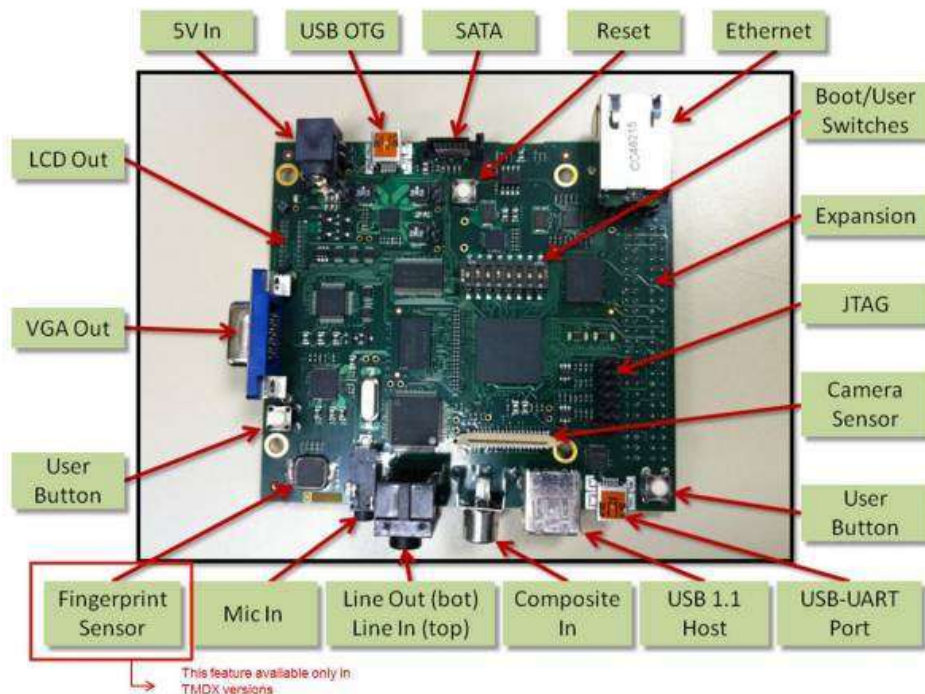
- One mini-USB Serial Port (on-board serial to USB)
- One Fast Ethernet Port (10/100 Mbps) with status LEDs
- One USB Host port (USB 1.1)
- One SATA Port (3Gbps)
- One LCD Port
- One Leopard Imaging Camera Sensor Input (36-pin ZIP connector)
- Three AUDIO Ports (1 LINE IN-J55 & 1 LINE OUT-J56 & 1 MIC IN-J57)
- 14-pin JTAG header (No onboard emulator; external emulator is required)

TMS320C6748 DSP Features

- ❖ Highest-Performance Floating-Point Digital Signal Processor (DSP):
 - Eight 32-Bit Instructions/Cycle
 - 32/64-Bit Data Word
 - 375/456-MHz C674x Fixed/Floating-Point
 - Up to 3648/2746 C674x MIPS/MFLOPS
 - Rich Peripheral Set, Optimized for Audio
 - Highly Optimized C/C++ Compiler
 - Extended Temperature Devices Available
- ❖ Advanced Very Long Instruction Word (VLIW) TMS320C67x™ DSP Core
 - Eight Independent Functional Units:
 - Two ALUs (Fixed-Point)
 - Four ALUs (Floating- and Fixed-Point)
 - Two Multipliers (Floating- and Fixed-Point)
 - Load-Store Architecture With 64 32-Bit General-Purpose Registers
 - Instruction Packing Reduces Code Size
 - All Instructions Conditional
- ❖ Instruction Set Features
 - Native Instructions for IEEE 754
 - Single- and Double-Precision
 - Byte-Addressable (8-, 16-, 32-Bit Data)
 - 8-Bit Overflow Protection
 - Saturation; Bit-Field Extract, Set, Clear; Bit-Counting; Normalization
- ❖ 67x cache memory.

- 32K-Byte L1P Program Cache (Direct-Mapped)
- 32K-Byte L1D Data Cache (2-Way)
- ❖ 256K-Byte L2 unified Memory RAM\Cache.
- ❖ Real-Time Clock With 32 KHz Oscillator and Separate Power Rail.
- ❖ Three 64-Bit General-Purpose Timers
 - Integrated Digital Audio Interface Transmitter (DIT) Supports:
 - S/PDIF, IEC60958-1, AES-3, CP-430 Formats
 - Up to 16 transmit pins
 - Enhanced Channel Status/User Data
 - Extensive Error Checking and Recovery
- ❖ Two Inter-Integrated Circuit Bus (I²C Bus™) .
- ❖ 3 64-Bit General-Purpose Timers (each configurable as 2 32-bit timers)
- ❖ Flexible Phase-Locked-Loop (PLL) Based Clock Generator Module
- ❖ IEEE-1149.1 (JTAG[†]) Boundary-Scan-Compatible
- ❖ 3.3-V I/Os, 1.2[‡]-V Internal (GDP & PYP)
- ❖ 3.3-V I/Os, 1.4-V Internal (GDP)(300 MHz only)
- ❖ Two Serial Peripheral Interfaces (SPI) Each With Multiple Chip-Selects
- ❖ Two Multimedia Card (MMC)/Secure Digital Card Interface with Secure Data I/O (SDIO) Interfaces
- ❖ One Multichannel Audio Serial Port.
- ❖ Two Multichannel Buffered Serial Ports

TOP VIEW OF TMS320C6748 LCDK Board



INSTALLATION

SYSTEM REQUIREMENTS:

	Recommended system requirements	Minimum system requirements
Processor	PENTIUM OR HIGHER	1.5GHz
RAM	4GB or above	1GB
Free Disk Space	4GB	4GB
Operating System	Microsoft windows 8/7/XP	-DO-

SOFTWARE INSTALLATION:

- CCSv6 or Higher(http://processors.wiki.ti.com/index.php/Download_CCS)
Version 6 or higher is recommended
- C64XPLUS-IMGLIB (<http://www.ti.com/tool/sprc264>)

HARDWARE REQUIREMENTS:

- Kit connected with 5V power supply
- USB cable connected with PC and JTAG emulator.
- Double sided stereo cable and speakers/ear phones (For Audio processing)
- Stereo cables, Function generator and CRO (For signal filtering applications)